

It starts from a set of short-range paths that the vehicle would be dynamically capable of completing, given its speed, direction and angular position, and removes all those that would either cross an obstacle or come too close to the predicted path of a moving vehicle. For example, a vehicle travelling at 80 kilometres per hour would not be able to safely complete a right turn five meters ahead, therefore that path would be eliminated from the feasible set.

Remaining paths are evaluated based on safety, speed and time requirements. Once the best path has been identified, a set of throttle, brake and steering commands are passed on to the vehicle's onboard processors and actuators. Altogether, this process takes on average 50 milliseconds. It can be longer or shorter depending on the amount of collected data, available processing power and complexity of the path-planning algorithm.

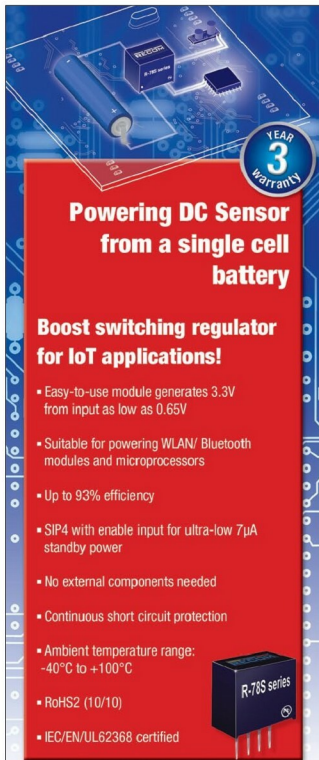
The process of localisation, mapping, obstacle detection and path planning is repeated until the vehicle reaches its destination. Currently, vehicle sensor data exists in multiple different formats across automakers. Pooling analogous vehicle data from millions of vehicles will be a key enabler for highly- and fully-automated driving, ensuring that each vehicle has a near-real-time view of road conditions and hazards, which can lead to better driving decisions.

The development is on for required location where Cloud technology that can detect and process changes in the real world as these happen, including on roads in dozens of countries, on an industrial scale and in high quality.

If a car around the next corner hits the brakes because there is an obstruction, that information could be used to signal to the drivers behind to slow down ahead of time, resulting in smoother, more efficient journeys and a lower risk of accidents. Standardised vehicle data exchange will enable the crowd-sourcing paradigm to spread across the fragmented automotive ecosystem, leveraging the synergies between connectivity and sensor data to provide smart mobility services such as real-time traffic, weather and parking spaces in the short term, while holding the promise to power self-driving cars with critical high-accuracy real-time mapping capabilities in the future.

ERTICO - ITS Europe includes in particular Advanced Driver Assistance Systems Interface Specifications (ADASIS), a forum that defines how maps connect and interact with the advanced driver assistance systems of a car.

Part 2 of the article will cover Lidar, GPS modules, MEMS devices and a distributed solution used for self-driving cars, next month.



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